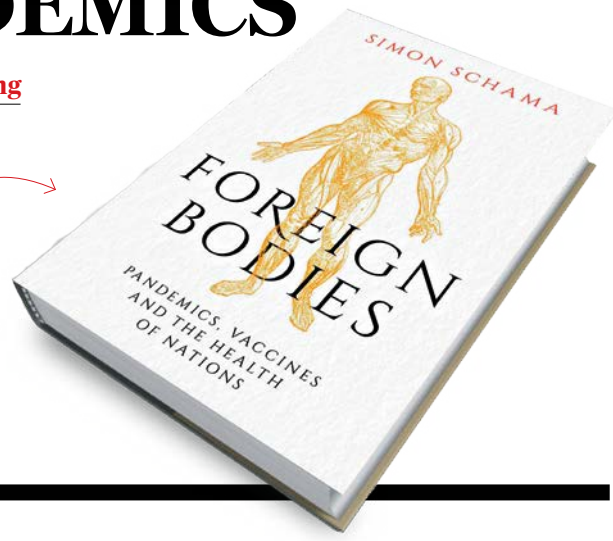
# A HISTORY OF PANDEMICS

By Gagandeep Kang




**FOREIGN BODIES**
**Pandemics, Vaccines and the Health of Nations**
*By Simon Schama*
SIMON & SCHUSTER
**₹899; 480 pages**

What comes to mind when we hear 'cholera'? Why is cholera still a problem in the world today? The Bay of Bengal has the dubious distinction of housing on its shores the only populations in the world where the disease has been endemic ('in the people') for centuries. In the 19th and 20th centuries, the movement of sailors, traders and indentured labour carried the infection to Europe, the Americas and Africa. The spread of cholera and the need for international cooperation was recognised by Adrien Proust, father of novelist Marcel Proust, before the causative agent was recognised in the late 19th century. Although Robert Koch is credited with the discovery of the cholera bacillus in 1884, John Snow had already demonstrated the role of contaminated water in transmission, Filippo Pacini had described the swarms of bacteria in intestinal fluids and Proust had mapped the spread of cholera.

These gripping tales of adventure, travel and insights and advancements in public health come at a time when the modern world has just dealt with a pandemic that is now the most studied infectious disease in history. Yet, we often forget that infectious diseases were the most common cause of death in any population anywhere throughout human history until a century ago.

*Foreign Bodies* brings together stories of diseases that decimate populations and the scientists past and present who studied them in a captivating journey through time, geography and evolving science in a little over 400 pages. Simon Schama's research into archives in the UK and Israel, and stitching together from secondary sources the tales of these diseases, illustrate the progress made in the understanding of the causes and spread of smallpox, cholera and plague, and the development of inoculation and then vaccines as tools for prevention of disease.

The book is organised in three main sections focused on these three diseases, although there are brief segues into other diseases to emphasise the importance of understanding the consequences of disturbing ecological niches, and how politics can play into public health decisions.

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The stories come together around the heroic and ultimately tragic figure of Waldemar Haffkine, whose experiences show the best possible impact and the worst consequences of the administration of science on the development and use of vaccines—a lesson that resonates today. We are told about his youth as an Odessa-born Jew who turned to activism and then science. The excitement of working with Elie Metchnikoff and Louis Pasteur and the frustration of dealing with the Indian Medical Service, which continued to treat him as an outsider, are well told. Throughout his travels, we hear of intersections and meetings with all the best scientists in the UK and Europe.

In India, he develops not one, but two vaccines, for cholera and for plague, and then takes them into the community for evaluation. In all of his work, Haffkine embodies the modern scientist, working with Indians as partners in the making and delivery of vaccines, concerned about safety, ensuring that there is careful measurement and the use of controls. But most British administrators are ready to find fault in his work and an unfortunate incident leads to his career being derailed until Ronald Ross comes to his rescue.

There are so many stories in this book that there could easily have been several deeper dives into the people and the science, and that is without getting into the remarkable history of the first large-scale vaccination campaigns in India. Well-researched, this book weaves together an intricate tapestry of the history of outbreaks and interventions, showing us how the world's linkages determine the health of our people and of the world. ■

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